CASE STUDY MANUAL

PROMPT ENGINEERING FOR GENERATIVE AI

TITLE: AI-Assisted Investigation and Verification of a Contemporary Global, National, Political, Technological, Economic, or Social Issue

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STUDENT INFORMATION

Field	Details
Student Name	
Enrollment Number	
Batch	
Department	
Date of Submission	
Selected Category	
Topic Title	

CASE STUDY OVERVIEW

In this case study, students will act as AI Research Analysts and Prompt Engineers. Students must independently identify a real-world issue and use Prompt Engineering techniques to investigate, verify, analyze, and report findings.

The objective is not to generate answers using AI, but to design prompts that improve reliability, minimize hallucinations, evaluate bias, and produce evidence-based conclusions.

LEARNING OUTCOMES

After completing this case study, students will be able to:

✓ Design effective prompts

✓ Apply Zero-shot and Few-shot Prompting

✓ Implement Role-based Prompting

✓ Apply Chain-of-Thought Prompting

✓ Use ReAct Prompting

✓ Design Prompt Chains

✓ Detect Hallucinations

✓ Evaluate Bias and Reliability

✓ Optimize Prompt Performance

✓ Produce Evidence-Based AI Reports

PHASE 1

TOPIC DISCOVERY AND APPROVAL

Step 1: Select Investigation Category

Tick One

Category	Selected
Government Policy	
Political Affairs	
International Relations	
Technology & AI	
Business & Economy	
Social Issues	
Historical Investigation	
Science & Research	

Step 2: Topic Proposal

Topic Title

Why did you select this topic?

Why may misinformation exist regarding this topic?

What challenges do you expect during verification?

Topic Approval Table

Parameter	**Details**
Topic Title	
Category	
Date Selected	
Faculty Approval	

PHASE 2

EVIDENCE COLLECTION

Source Collection Requirements

Source Type	Required	Collected
Official Sources	2	
News Articles	4	
Research Papers	2	
AI Responses	3	
Public Discussions	2	

Evidence Repository

Source No.	Source Type	Title	URL	Reliability (1-5)
1				
2				
3				
4				
5				

(Add additional rows)

PHASE 3

INITIAL AI INVESTIGATION

Prompt 1

Paste your first prompt.

____

AI Output Summary

Parameter	Observation
Main Findings	
Missing Information	
Unsupported Claims	
Potential Hallucinations	
Bias Observed	

Initial Evaluation Score

Criteria	Score (1-5)
Accuracy	

Criteria	Score (1-5)
Completeness	
Reliability	
Clarity	
Objectivity	

PHASE 4

ROLE PROMPTING

Prompt Used

Act as a Senior Investigative Journalist...

Paste complete prompt below.

Output Evaluation

Criteria	Initial Prompt	Role Prompt
Accuracy		
Depth		
Reliability		
Bias Control		
Evidence Usage		

Reflection

What improvements were observed?

PHASE 5

CHAIN OF THOUGHT ANALYSIS

Prompt Used

Paste Prompt

Source-Wise Claim Analysis

Source	Key Claims	Supporting Evidence
Source 1		
Source 2		
Source 3		

Contradiction Analysis

Claim	Source A	Source B	Conflict Found?
1			
2			
3			

PHASE 6

ReAct VERIFICATION

Prompt Used

Paste Prompt

Verification Matrix

Claim	Evidence Found	Verified?	Confidence (%)
1			
2			
3			

PHASE 7

HALLUCINATION DETECTION CHALLENGE

Task

Ask AI to provide:

- Additional facts
- Predictions
- Hidden causes
- Expert opinions

Hallucination Audit Table

AI Generated Statement	Evidence Found?	Verified?	Hallucination?
1			
2			
3			
4			

(Minimum 15 Entries)

Hallucination Summary

Parameter	Count
Total Claims Generated	
Verified Claims	

Parameter	Count
Unverified Claims	
Hallucinated Claims	

PHASE 8

PROMPT OPTIMIZATION

Prompt Evolution Record

Version 1

Prompt:

Weaknesses:

Version 2

Prompt:

Improvements:

Version 3

Prompt:

Improvements:

Version 4

Prompt:

Improvements:

Performance Comparison

Metric	V1	V2	V3	V4
Accuracy				
Reliability				
Completeness				
Hallucination Rate				
Bias Reduction				

PHASE 9

MULTI-LLM COMPARISON

Tools Used

☐ ChatGPT

☐ Gemini

☐ Claude

☐ Perplexity

☐ Copilot

Comparison Table

Criteria	Tool 1	Tool 2
Accuracy		
Reliability		
Hallucination Rate		
Bias		
Overall Performance		

PHASE 10

PROMPT CHAINING WORKFLOW

Investigation Workflow Diagram

Step	Objective	What Prompt Used to get details. (Mention your prompt)
1	Topic Understanding	
2	Source Extraction	
3	Claim Identification	
4	Contradiction Detection	
5	Fact Verification	
6	Bias Analysis	
7	Final Report Generation	

PHASE 11

ETHICS AND RELIABILITY ANALYSIS

Reliability Assessment

Source Type	Reliability (1-5)	Reason
Government Sources		
News Sources		
AI Outputs		
Social Media		

Bias Analysis

Source	Bias Type	AI Detected Bias?	Student Verification	Final Assessment
Source 1	Political	Yes	Verified	Moderate Bias
Source 2	Commercial	Yes	Verified	High Bias
Source 3	Ideological	No	Student Found Bias	Moderate Bias

Bias Reflection

1. Which source appeared most objective and why?

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2. Which source appeared most biased and why?

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3. Did the AI correctly identify all biases?

   ☐ Yes ☐ No

   If No, explain:

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4. How can prompt engineering reduce bias in AI-generated outputs?

Ethical Concerns Identified

PHASE 12

FINAL INVESTIGATION REPORT

Executive Summary

(300 Words)

Background

(300 Words)

Key Findings

(500 Words)

Verified Facts

Facts	Evidence Source
1	
2	

Identified Misinformation

Claims	Why Incorrect
1	
2	

Lessons Learned About Prompt Engineering

(300 Words)

Final Conclusion

(300 Words)

FACULTY EVALUATION RUBRIC

Criteria	Marks
Topic Selection	5
Evidence Collection	5
Prompt Design	10
CoT & ReAct Usage	10
Hallucination Detection	10
Prompt Optimization	10
Multi-LLM Comparison	5
Ethical Analysis	5
Final Report	15
Overall Professionalism	25
Total	100